Representing Inequalities on a Number Line

Exam code: 0607

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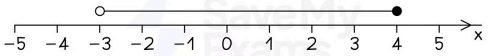
Representing inequalities on a number line

How do I represent an inequality on a number line?

- The inequality $-3 < x \leq 4$ is shown on a **number line** below



- Draw **circles** above the **end points** and connect them with a **horizontal line**
  - Leave an **open circle** for end points with **strict** inequalities, **< or >**
    - These end points **are not** included
  - Fill in a **solid circle** for end points with **≤ or ≥** inequalities
    - These end points **are** included

< or > use an open circle ○ (end points are excluded)
≤ or ≥ use a closed circle ● (end points are included)

- Use a **horizontal arrow** for inequalities with **one end point**
  - $x > 5$ is an open circle at 5 with a horizontal arrow pointing to the right

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Worked Example

Represent the following inequalities on a number line.

(a) $-2 \leq x < 1$

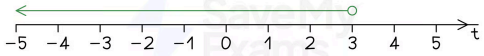
1 is not included so use an open circle

(b) $t < 3$

Answer:

3 is not included so use an open circle

There is no second end point
Any value less than three is accepted, so draw a horizontal arrow to the left



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